

MATHEMATICAL RESEARCH UPDATE

**IE-27: Continued analysis
and verified extensions**

The spectral disk for a Radau stage preconditioner

Sidney Holden

Center for Computational Biology, Flatiron Institute, Simons Foundation

Prepared with substantial AI assistance

Status: this archive is not a complete solution of IE-27. The unrestricted stage-number inequality is not proved here, and no counterexample to the stated real-positive-shift conjecture is supplied. The new results below do not remove that gap.

1 Target and scope

For the increasing Radau IIA nodes and their collocation matrix A_q , take

$$A_q^{-1} = L_q U_q, \quad \text{diag}(U_q) = \mathbf{1}, \quad B_q = L_q^{-1}, \quad N_q = U_q - I_q, \quad r_q = \|N_q\|_2 < 1.$$

The precise remaining target is

$$\boxed{\rho((I_q + \mu B_q)^{-1} N_q) \leq r_q} \quad (q \geq 2, r_q < 1, \mu \in \mathbb{R}_{>0}). \quad (1)$$

This is the small-matrix formulation of the requested disk enclosure for the symmetric positive-definite spatial pencil. It is given on the problem page and in the cited spectral reduction; the disk uses a *radius*, not a diameter [1, 2].

The prior nine-page report and its files are retained in `prior_results/`. That report gives the spatial reduction, exact Radau inverse formula, all-stage positive-pivot theorem, and exact two- and three-stage arguments. Its 64 certificates cover $q = 2, \dots, 64$ and $q = 80$.

This continuation adds an all-stage small/large-shift theorem, a rank-two weighted identity, a single-matrix sufficient condition for a stronger norm bound, and rigorous certificates for $q = 96$ and $q = 128$. It also gives exact counterexamples to two *broad* assertions that must not be substituted for the Radau conjecture.

Claim	Status in this archive
Equation (1) for every admissible stage number	Not established
All positive shifts for $q = 2, \dots, 64, 80, 96, 128$	66 exact certificates
Small and large positive shifts, every q	Proved in Section 3
Complex right-half-plane shifts in place of real shifts	False; exact example
Generic matrices in place of Radau matrices	False; exact example

2 All-stage structure and a rank-two identity

Let $0 < c_1 < \dots < c_q = 1$ be the specified nodes, and set

$$R(t) = \prod_{j=1}^q (t - c_j), \quad v_i = c_i R'(c_i), \quad V = \text{diag}(v_i), \quad X = \text{diag}(c_i).$$

The inverse differentiation formula, proved in the retained report, is

$$D_{ij} = \frac{v_i}{v_j(c_i - c_j)} \quad (i \neq j), \quad D_{ii} = \frac{1}{2c_i} \quad (i < q), \quad D_{qq} = \frac{q^2 + 1}{2}. \quad (2)$$

For completeness, this formula comes from the cardinal basis $t\ell_j(t)/c_j$ of polynomials of degree at most q vanishing at zero. Its derivative evaluations give A_q^{-1} . The diagonal simplification uses the equation

$$t(1-t)J''(t) + (1-3t)J'(t) + (q^2-1)J(t) = 0, \quad J(t) = P_{q-1}^{(1,0)}(2t-1),$$

whose roots are the interior nodes. The retained report supplies the coefficient-level derivation, so the certificate checker does not trust a rounded Butcher matrix.

Proposition 2.1. *For $C = V^{-1}DV$, $e = (1, \dots, 1)^T$, and the last coordinate vector e_q ,*

$$C + C^T = \text{diag}(1/c_1, \dots, 1/c_{q-1}, q^2 + 1) \succ 0, \quad (3)$$

$$XC + C^T X = ee^T + q^2 e_q e_q^T. \quad (4)$$

Every pivot of the exact no-pivot Crout factorization $D = LU$ is positive.

Proof. Off the diagonal, $C_{ij} = 1/(c_i - c_j)$, so (3) follows by cancellation. The off-diagonal entries of (4) are $(c_i - c_j)/(c_i - c_j) = 1$. Its first $q-1$ diagonal entries are one, and its last is $q^2 + 1$.

Every leading principal submatrix of C has positive-definite symmetric part. For each of its complex eigenpairs, the real part of the eigenvalue is positive, as follows by taking the Hermitian inner product with its eigenvector. Real eigenvalues are therefore positive, and conjugate pairs have positive product. Every leading determinant is positive. The same holds for D by diagonal similarity. Successive ratios of these leading determinants are exactly the pivots of the specified factorization. $\square$

Consequently $B = L^{-1}$ exists, has positive diagonal entries, and $I + \mu B$ is nonsingular for every real $\mu \geq 0$. These facts hold without first proving $r_q < 1$.

Identity (4) is useful additional structure, but it concerns D , not the Euclidean symmetric part of B . Replacing that distinction by an unproved accretivity assertion would invalidate a disk proof.

3 Two rigorously covered shift regimes at every stage

This result uses only the triangular structure and the positive pivots; it does not rely on a finite-stage computation.

Theorem 3.1. *For every $q \geq 2$, the requested enclosure holds whenever*

$$0 < \mu \leq \frac{1}{q \|B_q\|_2} \quad \text{or} \quad \mu \geq 2 \|L_q\|_2. \quad (5)$$

The second regime in fact gives the stronger inequality $\|(I + \mu B_q)^{-1} N_q\|_2 \leq r_q$.

Proof. Suppress the subscript q and write $r = \|N\|_2$. Here $r > 0$: for example, $U_{12} = D_{12}/L_{11} \neq 0$ by (2). Since N is strictly upper triangular, $N^q = 0$.

Suppose λ is an eigenvalue of $(I + \mu B)^{-1} N$ with $|\lambda| > r$, and choose a nonzero eigenvector x . Then

$$(\lambda I - N)x = -\lambda \mu Bx, \quad (\lambda I - N)^{-1} = \sum_{k=0}^{q-1} \frac{N^k}{\lambda^{k+1}}.$$

Taking norms gives

$$1 \leq \mu \|B\|_2 \sum_{k=0}^{q-1} \left(\frac{r}{|\lambda|} \right)^k < q \mu \|B\|_2.$$

This is impossible in the first regime, including its endpoint.

For the second regime, use

$$(I + \mu B)^{-1} N = (L + \mu I)^{-1} L N.$$

The Neumann-series estimate, valid for $\mu > \|L\|_2$, yields

$$\|(I + \mu B)^{-1} N\|_2 \leq \frac{\|L\|_2 r}{\mu - \|L\|_2} \leq r \quad \text{when } \mu \geq 2 \|L\|_2.$$

□

A sharper small-shift version replaces q by

$$\eta_q = \sum_{k=0}^{q-1} \frac{\|N_q^k\|_2}{r_q^k} \leq q.$$

The same argument works because the corresponding resolvent sum is strictly smaller than η_q when $|\lambda| > r_q$.

What this does not prove. For an uncertified stage, a possible violation is now confined to the finite interval

$$\frac{1}{q \|B_q\|_2} < \mu < 2 \|L_q\|_2.$$

The theorem does not cover that interval. Both its endpoints still depend on the stage, and their existence alone is not a uniform-stage solution.

4 A stronger norm criterion and the certificate criterion

Proposition 4.1. Fix q , write $B = L^{-1}$, $N = U - I$, $r = \|N\|_2 > 0$, and define

$$K = L \left(I - \frac{NN^T}{r^2} \right) L^T \succeq 0.$$

A sufficient condition for $\|(I + \mu B)^{-1}N\|_2 \leq r$ for every real $\mu \geq 0$ is the single inequality

$$L + L^T + 2K^{1/2} \succeq 0, \tag{6}$$

where $K^{1/2}$ is the symmetric positive-semidefinite square root.

Proof. Put $P = I + \mu B$. The desired norm inequality is equivalent, by congruence, to $r^2 PP^T - NN^T \succeq 0$. Congruence by L and division by r^2 turn this into

$$Q(\mu) = K + \mu(L + L^T) + \mu^2 I \succeq 0.$$

The exact identity

$$Q(\mu) = (\mu I - K^{1/2})^2 + \mu(L + L^T + 2K^{1/2})$$

proves the assertion. The first term is the square of a symmetric matrix; the second is positive semidefinite under (6). $\square$

This is a *sufficient* test. It has not been proved for all Radau stage counts. Failure of this stronger test would not by itself refute IE-27. In particular, the orientation PP^T and NN^T in this argument must not be interchanged with $P^T P$ and $N^T N$.

Proposition 4.2 (Common energy criterion). Suppose $H = H^T \succ 0$ and $t > 0$ satisfy

$$HB + B^T H \succeq 0, \quad tH - N^T H N \succeq 0.$$

Then $\rho((I + \mu B)^{-1}N) \leq \sqrt{t}$ for every real $\mu \geq 0$.

Proof. For $\|x\|_H = (x^* H x)^{1/2}$,

$$(I + \mu B)^T H (I + \mu B) = H + \mu(HB + B^T H) + \mu^2 B^T H B \succeq H.$$

Thus the inverse is nonexpansive in this norm, while $\|Nx\|_H \leq \sqrt{t}\|x\|_H$. The product has induced H -norm at most $\sqrt{t}$, which also bounds its spectral radius. $\square$

The exact certificates use Proposition 4.2; the numerical square-root experiment is *not* their proof mechanism.

5 Two additional exact certificates

The new files `new_certificates/q096.json` and `new_certificates/q128.json` pass the retained integer-interval verifier. They contain rational H, t, v and rational isolating intervals for every interior Radau node. The added squared radii are

$$t_{96} = \frac{7801459205}{10^{10}}, \quad t_{128} = \frac{8069729082}{10^{10}}.$$

For each of these stages, the verifier establishes

$$H \succ 0, \quad HB + B^T H \succ 0, \quad tH - N^T H N \succ 0, \quad I - N^T N \succ 0,$$

$$v \neq 0, \quad \|Nv\|_2^2 - t\|v\|_2^2 > 0.$$

The last two checks prove $\sqrt{t} < r < 1$. Combining them with Proposition 4.2 gives the *continuum* statement

$$\rho((I + \mu B_q)^{-1} N_q) \leq \sqrt{t_q} < r_q < 1 \quad \text{for every real } \mu > 0, \quad q \in \{96, 128\}.$$

Verified positive lower bound	$q = 96$	$q = 128$
Smallest L pivot	70.629700811196	94.174566164618
Smallest LDL pivot of H	1.001307579999	1.002030669999
Smallest LDL pivot of $HB + B^T H$	0.000156834663	0.000110365601
Smallest LDL pivot of $tH - N^T H N$	0.376277528400	0.436040730349
Smallest LDL pivot of $I - N^T N$	0.716423169363	0.716128872382
Rayleigh gap for supplied v	0.001562635417	0.001616369979

These are downward-rounded displays of certified rational lower bounds, not numerical eigenvalue estimates. LDL pivots are not eigenvalues.

Trust boundary and finite scope

Node brackets use 256 bits; all interval operations use integer endpoints with denominator 2^{1024} and outward rounding. Polynomial signs isolate the nodes; exact matrix formulas and triangular elimination enclose the factors; interval LDL elimination proves positivity. No floating-point optimizer or eigensolver is trusted by the verifier.

The complete supplied stage set is

$$\mathcal{S} = \{2, 3, \dots, 64\} \cup \{80, 96, 128\}, \quad |\mathcal{S}| = 66.$$

This does *not* mean all stages through 128. In particular, stages 65–79, 81–95, 97–127, and stages above 128 are not covered by these certificates. Existence of a certificate for every stage has not been proved.

From the extracted archive, `python code/run_all.py` runs the standard-library proof checks. Optional numerical generation is separate.

6 Why two tempting generic arguments are invalid

6.1 Positive pivots and weighted accretivity are insufficient

Consider the exact rational matrices

$$B = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}, \quad L = B^{-1} = \begin{pmatrix} 1 & 0 \\ -5 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1/10 \\ 0 & 0 \end{pmatrix}.$$

They have the stated triangular forms, positive pivots, and $\|N\|_2 = 1/10 < 1$. Moreover,

$$D = L(I + N) = \begin{pmatrix} 1 & 1/10 \\ -5 & 1/2 \end{pmatrix}, \quad H_0 = \text{diag}(50, 1),$$

$$H_0 D + D^T H_0 = \text{diag}(100, 1) \succ 0.$$

Nevertheless, at the positive real shift $\mu = 1$,

$$(I + B)^{-1}N = \begin{pmatrix} 0 & 1/20 \\ 0 & -1/8 \end{pmatrix}, \quad \rho((I + B)^{-1}N) = \frac{1}{8} > \frac{1}{10} = \|N\|_2.$$

The script `code/exact_obstructions.py` checks these identities and inequalities using rational arithmetic.

This is not a Radau matrix and is not a counterexample to IE-27. It proves that positive pivots, triangularity, and diagonal weighted accretivity of D do not suffice for a general argument. Any proof that uses only those properties has discarded essential structure.

6.2 The Euclidean inverse factor is not always nonexpansive

The recovered exact three-stage matrix is

$$B_3 = \begin{pmatrix} \frac{4}{5} - \frac{\sqrt{6}}{5} & 0 & 0 \\ \frac{9}{50} + \frac{29\sqrt{6}}{200} & \frac{3}{10} + \frac{3\sqrt{6}}{40} & 0 \\ \frac{4}{9} - \frac{\sqrt{6}}{36} & \frac{4}{9} + \frac{\sqrt{6}}{36} & \frac{1}{9} \end{pmatrix}.$$

For $w = (0, 1, -2)^T$,

$$w^T B_3 w = \frac{-52 + 7\sqrt{6}}{360} < 0.$$

Hence

$$\|(I + \mu B_3)w\|_2^2 - \|w\|_2^2 = 2\mu w^T B_3 w + \mu^2 \|B_3 w\|_2^2 < 0$$

for sufficiently small real $\mu > 0$. The inverse factor then has Euclidean norm greater than one. This invalidates the shortcut $\|(I + \mu B)^{-1}\|_2 \leq 1$, but does not invalidate the spectral bound for the product with N .

7 An exact failure for complex right-half-plane shifts

There is also a genuine Radau example outside the stated shift domain. For $q = 3$, take

$$\mu = \frac{1}{100} + 3i, \quad R_0 = \frac{41}{100}.$$

Exact arithmetic in $\mathbb{Q}(\sqrt{6}, i)$ establishes

$$\rho((I + \mu B_3)^{-1} N_3) > R_0 > \|N_3\|_2. \quad (7)$$

Here $\Re \mu > 0$, but μ is *not real*. Thus (7) does not refute IE-27; it prevents an unjustified extension of the target to the entire right half-plane.

Exact proof certificate

Let $s = \sqrt{6}$. Together with the B_3 on the previous page, use

$$N_3 = \begin{pmatrix} 0 & -\frac{53}{25} + \frac{76s}{75} & \frac{16}{25} - \frac{22s}{75} \\ 0 & 0 & \frac{2}{25} + \frac{3s}{25} \\ 0 & 0 & 0 \end{pmatrix}.$$

The checker first proves $R_0^2 I - N_3^T N_3 \succ 0$ by exact leading principal minors, establishing the second strict inequality in (7).

Set $Y = (I + \mu B_3)^{-1} N_3$. Its first column is zero. Its two nontrivial eigenvalues are the roots of $z^2 + az + b$, with

$$a = -Y_{22} - Y_{33}, \quad b = Y_{22}Y_{33} - Y_{23}Y_{32}.$$

Put $A = a/R_0$ and $E = b/R_0^2$. If both roots had modulus at most R_0 , then necessarily

$$|A - \overline{A}E| \leq 1 - |E|^2. \quad (8)$$

Indeed, writing the scaled roots as z_1, z_2 , with moduli $u, v \leq 1$,

$$A - \overline{A}E = -z_1(1 - |z_2|^2) - z_2(1 - |z_1|^2),$$

whose modulus is at most $(u + v)(1 - uv) \leq (1 + uv)(1 - uv) = 1 - |E|^2$.

The supplied checker obtains the two strict inequalities

$$1 - |E|^2 > 0, \quad |A - \overline{A}E|^2 - (1 - |E|^2)^2 > \frac{27}{100}.$$

They contradict (8). Every field operation is exact; signs are bounded using the checked rational bracket $2449489/10^6 < \sqrt{6} < 2449490/10^6$. Full rational components and bounds are in `results/exact_obstructions.json`; no numerical root-finding is part of this proof.

8 Numerical observations, reproducibility, and the gap

The optional script `code/diagnostics.py` evaluates the sufficient matrix inequality (6) and samples the shift family. Its output is explicitly marked `NUMERICAL_ONLY`. Selected estimates of the smallest eigenvalue of $L + L^T + 2K^{1/2}$ are

q	2	3	16	64	128
Numerical minimum	1.5943	0.7016	2.0536	2.1049	2.1238

The exact K is singular positive semidefinite; the numerical script clips negative roundoff before taking its square root. These numbers are neither exact positivity certificates nor an argument for untested stages. Likewise, maxima over the recorded 101-point shift grid are not maxima over all positive shifts. No conclusion depends on extrapolation from them.

Remaining mathematical obligation

The unrestricted statement (1) remains to be proved for $q \notin \mathcal{S}$ in the intermediate shift regime of Section 3, or refuted by a counterexample satisfying *all* of its hypotheses. One sufficient route would be a proof of (6) for all stages. Another would be a construction of the common-energy certificates for every stage with $t \leq r_q^2$. Neither existence result is established here. The finite certificate generator is not known to succeed for every stage and is not presented as such a proof.

Files and independent checks

The top-level report is this continuation; `prior_results/report.pdf` is the recovered earlier manuscript. The new exact checks are in `code/`, the added certificates in `new_certificates/`, and machine-readable outcomes in `results/`. Run

```
python code/run_all.py
```

for all 66 supplied certificates, the exact low-stage checks, and the regression tests. Run `python code/exact_obstructions.py` to check only the two rejected extensions. Verification requires only the Python standard library. Optional candidate generation and numerical diagnostics require NumPy and SciPy. The SHA-256 manifest checks file integrity, not mathematical truth.

References

- [1] *OpenProblemsInNLA* repository, IE-27, “Spectral disk for a Radau stage preconditioner.” Problem statement checked 12 September 2026; page status dated 11 September 2026.
<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination/IE-27>
- [2] M. Outrata, *On the recent advances of spectral analysis for systems arising from fully-implicit RK methods*, arXiv:2510.21241v1 (2025), Sections 2.1–2.3.
<https://arxiv.org/html/2510.21241v1>